

Spatial Transcriptomics Cheatsheet

The tables below consist of valuable functions or commands that will help you through this module.

Each table represents a different library/tool and its corresponding commands.

Please note that these tables are not intended to tell you all the information you need to know about each command.

The hyperlinks found in each piece of code will take you to the documentation for further information on the usage of each command. Please be aware that the documentation will generally provide information about the given function's most current version (or a recent version, depending on how often the documentation site is updated). This will usually (but not always!) match what you have installed on your machine. If you have a different version of R or other R packages, the documentation may differ from what you have installed.

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SpatialExperiment

Read the [SpatialExperiment package documentation \(and e-book\)](#) and a [vignette on its usage](#).

`SpatialExperiment` objects are based on `SingleCellExperiment` objects. There are many functions that are shared between the two, with equivalent usage. See [this SingleCellExperiment vignette](#) for example usage of shared functions. You can also refer to the Data Lab cheatsheets for our "Introduction to scRNA-seq" (`scRNA-seq-cheatsheet.pdf`) and "Advanced scRNA-seq" (`scRNA-seq-advanced-cheatsheet.pdf`) training modules.

Library/Package	Piece of Code	What it's called	What it does
<code>SpatialExperiment</code>	<code>SpatialExperiment()</code>	Spatial Experiment	Creates a <code>SpatialExperiment</code> object
<code>SpatialExperiment</code>	<code>spatialCoords()</code>	Spatial Experiment coordinates	Extracts spatial coordinates of spots in the <code>SpatialExperiment</code> object
<code>SpatialExperiment</code>	<code>imgData()</code>	Spatial Experiment images	View the image data associated with a <code>SpatialExperiment</code> object

VisiumIO

Read the R package [VisiumIO documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
<code>VisiumIO</code>	<code>TENxVisium()</code>	10x Visium	Constructor function for importing 10x Visium data
<code>VisiumIO</code>	<code>import()</code>	Import	Loads 10x Visium data defined with the <code>TENxVisium()</code> constructor function

SpatialExperimentIO

Read the [SpatialExperimentIO package documentation](#) and a [vignette on its usage](#).

This package provides reader functions for imaging-based spatial transcriptomics platforms, including Xenium (10x Genomics), CosMx (NanoString), MERSCOPE (Vizgen), and STARmap PLUS.

Library/Package	Piece of Code	What it's called	What it does
<code>SpatialExperimentIO</code>	<code>readXeniumSXE()</code>	Read Xenium into a Spatial Experiment	Loads an unzipped 10x Genomics Xenium output bundle directory into a <code>SpatialExperiment</code> object

SpatialFeatureExperiment

Read the [SpatialFeatureExperiment](#) package documentation.

This package provides functions for working with spatial transcriptomics data in the `SpatialFeatureExperiment` format, including reading data from various platforms (Visium, CosMx, Xenium), transforming spatial coordinates, and constructing spatial graphs.

Library/Package	Piece of Code	What it's called	What it does
<code>SpatialFeatureExperiment</code>	<code>toSpatialFeatureExperiment(x)</code>	Convert to SFE	Convert a <code>SpatialExperiment</code> or <code>Seurat</code> object to a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>read10xVisiumSFE()</code>	Read 10x Visium	Read 10x Visium data from Space Ranger into a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>readXenium()</code>	Read Xenium	Read Xenium data into a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>mirror()</code>	Mirror	Flip the spatial coordinates of a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>mirrorImg()</code>	Mirror image	Flip the image in a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>colGraph()</code>	Column graph	Get or set the column graph of a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>bbox()</code>	Bounding box	Get the bounding box (crop) of a <code>SpatialFeatureExperiment</code> object
<code>SpatialFeatureExperiment</code>	<code>findDebrisCells()</code>	Find debris cells	Identify debris cells in a <code>SpatialFeatureExperiment</code> object (or the spatial coordinates of the cells)
<code>SpatialFeatureExperiment</code>	<code>findSpatialNeighbors()</code>	Spatial neighbor graph	Create a graph representing the spatial relationships between spots
<code>SpatialFeatureExperiment</code>	<code>findVisiumGraph()</code>	Visium graph	Create a graph representing the spatial relationships between spots in a Visium dataset

ggspavis

Read the [ggspavis package documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
ggspavis	<code>plotCoords()</code>	Plot coordinates	Creates a spot plot showing spatial locations in x/y with options for spot annotation
ggspavis	<code>plotVisium()</code>	Plot Visium	Creates a spot plot for Visium data specifically showing spatial locations in x/y with options for spot annotation as well as an option to show the spatial image, e.g., the H&E image

Voyager

Read the [Voyager package documentation](#).

Library/Package	Piece of Code	What it's called	What it does
Voyager	<code>runMoransI()</code>	Moran's I	Computes Moran's I statistic for spatial autocorrelation in a <code>SpatialFeatureExperiment</code> object
Voyager	<code>runUnivariate()</code>	Univariate analysis	Performs univariate analysis for spatial features in a <code>SpatialFeatureExperiment</code> object using various methods (including Moran's I tests)
Voyager	<code>plotImage()</code>	Plot image	Plots an image (e.g., H&E image) associated with a <code>SpatialFeatureExperiment</code> object
Voyager	<code>plotColGraph()</code>	Plot column graph	Performs multivariate analysis for spatial features in a <code>SpatialFeatureExperiment</code> object using various methods
Voyager	<code>plotSpatialFeature()</code>	Plot spatial feature	Plots a feature value using the spatial coordinates in a <code>SpatialFeatureExperiment</code> object

SpotSweeper

Read the [SpotSweeper package documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
SpotSweeper	<code>localOutliers()</code>	Local outliers	Detect local outlier spots in a <code>SpatialExperiment</code> based on a given quality control metric
SpotSweeper	<code>plotQCMetrics()</code>	Plot QC metrics	Create a spot plot highlighting outliers detected with <code>SpotSweeper::localOutliers()</code>

Banksy

Read the [Banksy package documentation](#), which contains several vignettes on its usage, including a [vignette on parameter selection](#). Note that many of their vignettes use of image-based spatial transcriptomics data that is more fine-grained compared to 10x Visium.

Library/Package	Piece of Code	What it's called	What it does
Banksy	<code>computeBanksy()</code>	Compute <code>Banksy</code>	Compute the component neighborhood matrices for the <code>Banksy</code> matrix
Banksy	<code>runBanksyPCA()</code>	Run <code>Banksy</code> PCA	Run PCA on a <code>Banksy</code> matrix with a weighted spatial neighborhood context

spacexr

Read the [spacexr package documentation](#), and a [vignette on using their RCTD algorithm](#).

Library/Package	Piece of Code	What it's called	What it does
spacexr	<code>createRctd()</code>	Create an RCTD object	Preprocess data before performing deconvolution with RCTD
spacexr	<code>runRCTD()</code>	Run RCTD	Run the RCTD algorithm to decompose cell type mixtures
spacexr	<code>plotCellTypeWeight()</code>	Plot a single cell type's weights	Plot pixel proportions for a specific cell type on a spatial coordinates plot

hoodscanR

Read the [hoodscanR package documentation](#) and a [vignette on its usage](#).

Note that the functions in this package are generally run in sequence, where the output of one function is the input to the next.

Library/Package	Piece of Code	What it's called	What it does
hoodscanR	<code>findNearCells()</code>	Find the k-the nearest cells for each	Find the <code>k</code> nearest cells to each cell in a <code>SpatialExperiment</code> object, returning a list with a matrix of distances and a matrix of the annotations (e.g., cell types) of those neighboring cells
hoodscanR	<code>scanHoods()</code>	Scan cellular neighborhoods	Apply a modified softmax algorithm to the distance matrix from <code>hoodscanR::findNearCells()</code> to calculate the probability that each cell is associated with each of its nearest neighboring cells
hoodscanR	<code>mergeByGroup()</code>	Merge probability matrix based on annotations	Collapse the probability matrix from <code>hoodscanR::scanHoods()</code> by annotation, so that each column is a group (e.g., a cell type) rather than an individual neighboring cell
hoodscanR	<code>mergeHoodSpe()</code>	Merge probability matrix into Spatial Experiment object	Add the merged probability matrix into the <code>colData</code> of a <code>SpatialExperiment</code> object
hoodscanR	<code>calcMetrics()</code>	Calculate metrics for probability matrix	Calculate the entropy and perplexity of the probability matrix for each cell, which summarize how mixed or how distinct that cell's neighborhood is, and store them in <code>colData</code>
hoodscanR	<code>plotColocal()</code>	Plot heatmap for neighborhood analysis	Plot a heatmap of the Pearson correlations between cells' neighborhood probability distributions to show which groups tend to colocalize
hoodscanR	<code>clustByHood()</code>	Cluster the probability matrix with K-means	Perform k-means clustering on the probability matrix to assign each cell to a neighborhood cluster
hoodscanR	<code>plotProbDist()</code>	Plot probability distribution	Plot the distribution of neighborhood probabilities, optionally split by a grouping variable such as the neighborhood clusters from <code>hoodscanR::clustByHood()</code>

scuttle, scran, and scater

Read the [scuttle package documentation](#), and a [vignette on its usage](#).

Read the [scran package documentation](#), and a [vignette on its usage](#).

Read the [scater package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
scuttle	<code>addPerCellQC()</code>	Add per cell quality control	For a <code>SingleCellExperiment</code> object, calculate and add quality control per cell and store in <code>colData</code>
scuttle	<code>computeLibraryFactors()</code>	Compute Library Factors	Returns a numeric vector of computed size factors for each spot (or cell) stored in a <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code>) object. The size factor is computed as the library size of each spot/cell after scaling them to have a mean of 1 across all spots/cells
scuttle	<code>logNormCounts()</code>	Normalize log counts	Returns the <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code>) object with normalized expression values for each spot (cell), using the size factors stored in the object
scuttle	<code>isOutlier()</code>	Identify outliers	Convenience function to determine which values in a numeric vector are outliers based on the median absolute deviation (MAD)
scran	<code>getTopHVGs()</code>	Get top highly variable genes	Identify variable genes in a <code>SingleCellExperiment</code> object, based on variance
scran	<code>modelGeneVar()</code>	model per gene variance	Model the per gene variance of a <code>SingleCellExperiment</code> object
scran	<code>clusterCells()</code>	Cluster cells	Perform clustering on an SCE object using the <code>bluster</code> package
scater	<code>runPCA()</code>	Run PCA	Calculates principal components analysis on a <code>SingleCellExperiment</code> object, returning an SCE object with a PCA reduced dimension
scater	<code>runUMAP()</code>	Run UMAP	Calculates uniform manifold approximate projection on a <code>SingleCellExperiment</code> object, returning an SCE object with a UMAP reduced dimension
scater	<code>plotUMAP()</code>	Plot UMAP	Plot the "UMAP"-named reduced dimension slot from a <code>SingleCellExperiment</code> object

bluster

Read the [bluster package documentation](#) and vignettes on its usage:

- [Flexible clustering for Bioconductor](#)
- [Assorted clustering diagnostics](#)

Library/Package	Piece of Code	What it's called	What it does
bluster	<code>NNGraphParam()</code>	Graph-based clustering parameters	Set up parameters for nearest-neighbor (NN) graph-based clustering algorithms within <code>scrn::clusterCells()</code> or <code>bluster::clusterRows()</code>
bluster	<code>approxSilhouette()</code>	Approximate silhouette width	Calculate an approximate silhouette width for each cell given a set of clusters
bluster	<code>neighborPurity()</code>	Compute neighborhood purity	Calculate neighborhood purity for each cell given a set of clusters
bluster	<code>bootstrapStability()</code>	Assess cluster stability by bootstrapping	Generate cluster bootstrap replicates to estimate cluster robustness to sampling noise
bluster	<code>pairwiseRand()</code>	Compute pairwise Rand indices	Calculate the Rand index (adjusted, by default) between pairs of clustering results, using one of several available calculation modes

patchwork

Read the [patchwork package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
patchwork	<code>+</code>	Plot arithmetic	Place <code>ggplot2</code> objects side-by-side by adding them together with a <code>+</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots
patchwork	<code>/</code>	Plot arithmetic	Stack <code>ggplot2</code> objects on top of one another "dividing" them with a <code>/</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots
patchwork	<code>wrap_plots()</code>	Wrap plots	Wrap multiple <code>ggplot2</code> objects into a single multi-panel plot
patchwork	<code>plot_layout()</code>	Customize layout	Customize the layout of plots arranged with <code>patchwork</code>

pals

Read the [pals package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
pals	<code>alphabet()</code>	Alphabet palette	Create a discrete palette from the "alphabet" set of colors

purrr

Read the [purrr package documentation](#) and a [vignette on its usage](#), and download the [purrr package cheatsheet](#).

Library/Package	Piece of Code	What it's called	What it does
purrr	<code>map()</code>	map	Apply a function across each element of list; return a list
purrr	<code>imap()</code>	imap	Apply a function across each element of list and its index/names; return a list
purrr	<code>reduce()</code>	Reduce	Reduce a list to a single value by repeatedly applying a given function. Can also be used to iteratively modify a single object.

pheatmap

Read the [pheatmap package documentation](#).

Library/Package	Piece of Code	What it's called	What it does
pheatmap	<code>pheatmap()</code>	Pretty heatmap	Plot a (pretty!) clustered heatmap

ComplexHeatmap

Read the [ComplexHeatmap package documentation](#) and the [ComplexHeatmap Complete Reference e-book](#).

Library/Package	Piece of Code	What it's called	What it does
ComplexHeatmap	<code>Heatmap()</code>	Heatmap	Create a clustered heatmap from a matrix, with options to split, annotate, and combine it with other heatmaps